Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2003-II-Q03] | Part A (Q1-Q5)

Answer: 192

1. The first position has three choices. After any letter, exactly one successor is forbidden and two are allowed.

2. The count is $3 \cdot 2^6 = 192$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2004-II-Q04] | Part A (Q1-Q5)

Answer: 927

1. All 99 one- or two-digit numbers qualify. At length $L \geq 3$, either all digits agree, giving nine numbers, or choose a nonzero leading digit a and a distinct digit b: $9 \cdot 9$ choices.

2. The later $L-1$ positions use a, b, with at least one b, giving $2^{L-1}-1$ patterns. Hence the three- and four-digit counts are $9+81 \cdot 3=252$ and $9+81 \cdot 7=576$.

3. Total $99+252+576=927$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2005-I-Q05] | Part A (Q1-Q5)

Answer: 630

1. Read orientations bottom to top as U, D. The forbidden adjacency is UD, so every valid string consists of some down-facing coins followed by up-facing coins: nine choices for the split.

2. Independently choose the four gold positions in $\binom{8}{4} = 70$ ways. Total $9 \cdot 70 = 630$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2006-II-Q04] | Part A (Q1-Q5)

Answer: 462

1. The valley value a_6 is less than every other entry, so it is one. Choose the five numbers before it from the other eleven.

2. Their decreasing order and the remaining six numbers increasing order are forced. The number is $\binom{11}{5} = 462$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2007-II-Q01] | Part A (Q1-Q5)

Answer: 372

1. Strings with at most one zero use five distinct symbols from seven, giving $(7)_5 = 2520$.

2. With two zeroes, choose their positions in $\binom{5}{2}$ ways and order three distinct remaining symbols in $(6)_3$ ways, giving 1200. Thus $N = 3720$, so $N/10 = 372$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2011-I-Q05] | Part A (Q1-Q5)

Answer: 144

1. Subtract successive three-label sums: labels three places apart have the same residue modulo 3. Thus each of the three position classes modulo 3 carries one residue class.

2. The residue classes may be assigned in $3!$ ways, and their digits arranged in $(3!)^3$ ways. Every resulting triple has residues $0, 1, 2$, so all $6^4 = 1296$ labelings qualify.

3. Distinct digits prevent nonidentity rotational symmetries, so each orbit has size nine. The answer is $1296/9 = 144$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2012-I-Q03] | Part A (Q1-Q5)

Answer: 216

1. Choose the unique correctly served diner in nine ways. Suppose this is a beef orderer. Two beef meals remain for three chicken and three fish orderers, with the remaining two beef orderers receiving no beef.

2. If both leftover beef meals go to one group, choose their recipients in $2\binom{3}{2} = 6$ ways; all other types are forced. If they split between groups, there are $3 \cdot 3 = 9$ recipient choices, followed by two ways to assign the final chicken and fish meals to the two beef orderers.

3. The count is $9(6 + 9 \cdot 2) = 216$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2019-I-Q04] | Part A (Q1-Q5)

Answer: 122

1. There are always eleven choices of outgoing player. Incoming choices decrease from eleven to ten to nine. Thus the count is $1 + 11 \cdot 11 + (11 \cdot 11)(11 \cdot 10) + (11 \cdot 11)(11 \cdot 10)(11 \cdot 9)$.

2. This is 1331122, with remainder 122.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2019-II-Q05] | Part A (Q1-Q5)

Answer: 520

1. View the six even chairs as a cycle, with odd chairs between them. For a block of j occupied even chairs, advisors can choose adjacent odd chairs in $j+1$ ways: a string of left choices followed by right choices; a right-then-left boundary would collide.

2. Classify the two empty even chairs: adjacent (six choices) leaves one block of four, weight five; one chair between (six choices) gives blocks one and three, weight eight; opposite (three choices) gives blocks two and two, weight nine.

3. Total group-seat patterns $6 \cdot 5 + 6 \cdot 8 + 3 \cdot 9 = 105$. Assign the ambassador-advisor pairs in $4!$ ways, obtaining 2520, remainder 520.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2020-I-Q05] | Part A (Q1-Q5)

Answer: 052

1. For increasing remainders, insert a chosen exceptional label into the increasing order of the other five: $6 \cdot 6 = 36$ constructions.

2. The increasing row is counted six times, and each of the five adjacent swaps twice. No other duplication is possible: two possible deleted labels would have to form the only inverted pair, hence be adjacent. The distinct count is $36 - 5 - 5 = 26$.

3. Reversal gives another 26. An increasing and a decreasing subsequence of length five in six distinct labels would overlap in at least four labels, impossible since two such subsequences share at most one label. Total 52.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2021-II-Q03] | Part A (Q1-Q5)

Answer: 080

1. Fix the position i occupied by 3. Exactly two summands avoid it; modulo 3 their sum is $x_{i+2}x_{i+3}(x_{i+1}+x_{i+4})$.

2. The product factor is nonzero, so the two cyclic neighbors of 3 must lie in opposite nonzero residue classes. There are two choices per class and two orders, giving 8 neighbor assignments.

3. The other two labels have 2 orders, and 3 has 5 positions. Total $5\cdot 8\cdot 2=80$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2025-I-Q03] | Part A (Q1-Q5)

Answer: 016

1. Positive strictly decreasing counts summing to 9 are $(6,2,1),(5,3,1),(4,3,2)$.

2. They contribute multinomial counts $9!/(6!2!)=252$, $9!/(5!3!)=504$, and $9!/(4!3!2!)=1260$. The total is 2016, with remainder 16.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2004-I-Q06] | Part B (Q6-Q10)

Answer: 882

1. For any four distinct ordered digit values, five permutations have this alternating pattern. Place the largest in position two: three choices for position one, with the last pair forced. Or place it in position four: two choices for the remaining low positions.

2. If zero is absent, all five work, contributing $5\binom{9}{4} = 630$. If zero is present, it must be in position one or three; only three survive with zero in position three, contributing $3\binom{9}{3} = 252$.

3. Total 882.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2023-II-Q07] | Part B (Q6-Q10)

Answer: 928

1. Any inscribed rectangle has two diameters as its diagonals. Group the vertices into six opposite pairs; a forbidden rectangle occurs precisely when two pairs are both solid red or two pairs are both solid blue.

2. So there may be at most one all-red pair and at most one all-blue pair. Every remaining pair is mixed, with two orientations.

3. The four cases (zero solids, red only, blue only, both) give $2^6 + 6 \cdot 2^5 + 6 \cdot 2^5 + 6 \cdot 5 \cdot 2^4 = 928$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2025-I-Q10] | Part B (Q6-Q10)

Answer: 081

1. Fill the first block in $9!$ ways and label its row-digit sets R_1, R_2, R_3. The middle-block row sets partition the digits and row i must avoid R_i.

2. If its first row uses j members of R_2 and $3-j$ of R_3, the choices for the first and second row contents total $\binom{3}{j}^3$. Summing over $j=0,1,2,3$ gives 56. All third-block row contents are then forced.

3. The six row-triples in the latter two blocks may each be internally ordered in $3!$ ways. Thus the number is $9! \cdot 56 \cdot 6^6 = 2^{16}3^{10}5 \cdot 7^2$, giving $32+30+5+14=81$.

Source / 来源

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2026-I-Q07] | Part B (Q6-Q10)

Answer: 396

1. Every onto self-map of a finite set is a permutation. The sixth iterate is identity precisely when all cycle lengths divide six, so only lengths four and five are forbidden.

2. The excluded cycle types $4+1+1, 4+2, 5+1$ have counts $6!/(4 \cdot 2!) = 90$, $6!/(4 \cdot 2) = 90$, and $6!/5 = 144$. Thus the count is $720 - 324 = 396$.

Source / 来源